

APO-TAMSULOSIN CR TABLETS 0.4mg

Tamsulosin Hydrochloride 0.4mg

Product Description

Yellowish-brown, round, biconvex film-coated tablet. Engraved "TA" over "0.4" on one side, plain on the other side.

Pharmacodynamics

Mechanism of Action

Tamsulosin binds selectively and competitively to postsynaptic α_1 -receptors, in particular to the subtype α_{1A} , which bring about relaxation of the smooth muscle of the prostate, whereby tension is reduced.

Pharmacodynamic:

Tamsulosin increases maximum urinary flow rate by reducing smooth muscle tension in the prostate and urethra thereby, relieving obstruction.

It also improves the complex of irritative and obstructive symptoms in which bladder instability and tension of the smooth muscles of the lower urinary tract play an important role. α_1 -blockers can reduce blood pressure by lowering peripheral resistance. No reduction in blood pressure of any clinical significance was observed during studies with Tamsulosin.

Paediatric Population: No dose response was observed for any dose level.

Pharmacokinetics

Absorption: Tamsulosin is a controlled-release tablet and is absorbed from the intestine. Under fasting conditions, approximately 57% of the administered dose is estimated to be absorbed. A consistent control release of tamsulosin is maintained over the whole pH range encountered in the gastrointestinal tract with little fluctuation over 24 hrs. The extent of absorption is increased by 64% and 149% [area under the concentration-time curve (AUC) and peak plasma concentration (C_{max})].

Tamsulosin shows linear kinetics.

After a single dose of Tamsulosin in the fasted state, plasma levels of tamsulosin peak at a median time of 6 hrs. In steady state, which is reached by day 4 of multiple dosing, plasma levels of tamsulosin peak at 4-6 hrs in the fasted and fed state. Peak plasma levels increase from approximately 6 ng/mL after the 1st dose to 11 ng/mL in steady state.

As a result of the prolonged release characteristics of Tamsulosin controlled-release tablets, the trough concentration of tamsulosin in plasma amounts to 40% of the peak plasma concentration under fasted and fed conditions.

There is a considerable interpatient variation in plasma levels, both after single and multiple dosing.

Distribution: In man, tamsulosin is about 99% bound to plasma proteins and volume of distribution is small (about 0.2 L/kg).

Metabolism: Tamsulosin has a low first-pass effect, being metabolized slowly. Most tamsulosin is present in plasma in the form of unchanged drug. It is metabolised in the liver.

In rats, hardly any induction of microsomal liver enzymes was seen to be caused by tamsulosin.

No dose adjustment is warranted in hepatic insufficiency.

None of the metabolites are more active than the original compound.

Elimination: Tamsulosin and its metabolites are mainly excreted in the urine. Following intravenous or oral administration of an immediate-release formulation, the elimination half-life of

tamsulosin in plasma ranged from 5-7 hrs. Because of absorption rate-controlled pharmacokinetics with the tamsulosin hydrochloride controlled-release tablets formulation, the apparent half-life of tamsulosin increases to approximately 12-15 hrs in healthy volunteers.

No dosage adjustment is necessary in patients with renal impairment.

Toxicology: The general toxicity profile, as seen with high doses of tamsulosin, is consistent with the known pharmacological actions of the α -adrenergic blocking agents. At very high dose levels, the electrocardiogram (ECG) was altered in dogs. This response is considered to be not clinically relevant. Tamsulosin showed no relevant genotoxic properties.

Increased incidences of proliferative changes of mammary glands of female rats and mice have been reported. These findings, which are probably mediated by hyperprolactinaemia and only occurred at high dose levels, are regarded as irrelevant.

Indication

Treatment of functional symptoms of benign prostatic hyperplasia (BPH).

Recommended Dosage

1 tab daily. Apo-Tamsulosin CR Tablets should be taken at the same time each day with or without food.

Administration: Apo-Tamsulosin CR Tablets must be swallowed whole, and should not be crunched or chewed as this will interfere with the controlled release of the active ingredient.

Mode of Administration

Oral

Contraindications

Hypersensitivity to tamsulosin hydrochloride or to any other component of Apo-Tamsulosin CR Tablets.

History of orthostatic hypotension; severe hepatic insufficiency.

Warnings and Precautions

As with other α_1 -blockers, a reduction in blood pressure can occur in individual cases during treatment with Apo-Tamsulosin CR Tablets, as a result of which, rarely, syncope can occur. At the 1st signs of orthostatic hypotension (dizziness, weakness), the patient should sit or lie down until the symptoms have disappeared.

Before therapy with Apo-Tamsulosin CR Tablets is initiated, the patient should be examined in order to exclude the presence of other conditions which can cause the same symptoms as benign prostatic hyperplasia. Digital rectal examination and when necessary, determination of prostate specific antigen (PSA) should be performed before treatment and at regular intervals afterwards.

The treatment of severely renally impaired patients (creatinine clearance <10 mL/min) should be approached with caution as these patients have not been studied.

The 'intraoperative floppy iris syndrome' (IFIS, a variant of small pupil syndrome) has been observed during cataract surgery in some patients on or previously treated with tamsulosin. Intraoperative floppy iris syndrome may lead to increased procedural complications during the operation.

Discontinuing tamsulosin 1-2 weeks prior to cataract surgery is anecdotally considered helpful, but the benefit of treatment discontinuation has not been established. IFIS has also been reported in patients who had discontinued tamsulosin for a longer period prior to cataract surgery.

The initiation of therapy with tamsulosin hydrochloride in patients for whom cataract surgery is scheduled is not recommended. During pre-operative assessment, cataract surgeons and ophthalmic teams

should consider whether patients scheduled for cataract surgery are being or have been treated with tamsulosin in order to ensure that appropriate measures will be in place to manage the IFIS during surgery.

Tamsulosin hydrochloride should not be given in combination with strong inhibitors of CYP3A4 (eg, ketoconazole) in patients with poor metabolizer CYP2D6 phenotype.

It should be used with caution in combination with strong (eg, ketoconazole) and moderate (eg, erythromycin) inhibitors of CYP3A4 (see Interactions).

Effects on the Ability to Drive or Operate Machinery: No data is available on whether Apo-Tamsulosin CR Tablets adversely affects the ability to drive or operate machines. However, in this respect, patients should be aware of the fact that drowsiness, blurred vision, dizziness and syncope can occur.

Use in children: The safety and efficacy of tamsulosin in children <18 years have not been established.

Interactions with Other Medicaments

No interactions have been seen when tamsulosin was given concomitantly with atenolol, enalapril, nifedipine or theophylline. Concomitant cimetidine brings about a rise in plasma levels of tamsulosin and furosemide a fall, but as levels remain within the normal range, posology need not be changed.

In vitro, neither diazepam nor propranolol, trichlormethiazide, chlormadinon, amitriptyline, diclofenac, glibenclamide, simvastatin and warfarin change the free fraction of tamsulosin in human plasma. Neither does tamsulosin change the free fractions of diazepam, propranolol, trichlormethiazide and chlormadinon.

Diclofenac and warfarin, however, may increase the elimination rate of tamsulosin.

Concomitant administration of tamsulosin hydrochloride with paroxetine, a strong inhibitor of CYP2D6, resulted in a C_{max} and AUC of tamsulosin that had increased by a factor of 1.3 and 1.6, respectively, but these increases are not considered clinically relevant.

There is a theoretical risk of enhanced hypotensive effect when given concurrently with drugs which may reduce blood pressure, including anaesthetic agents and other α_1 -adrenoceptor antagonists.

Pregnancy and Lactation

Not applicable, as Apo-Tamsulosin CR Tablets is intended for male patients only.

Adverse Effects / Undesirable Effects

Frequencies are defined as: Common (>1/100, <1/10), uncommon (>1/1000, <1/100), rare (>1/10,000, <1/1000) and very rare (<1/10,000).

Nervous System Disorders: Common: Dizziness. Uncommon: Headache. Rare: Syncope.

Cardiac Disorders: Uncommon: Palpitations.

Vascular Disorders: Uncommon: Orthostatic hypotension.

Respiratory, Thoracic and Mediastinal Disorders: Uncommon: Rhinitis.

Gastrointestinal Disorders: Uncommon: Constipation, diarrhea, nausea, vomiting.

Skin and Subcutaneous Tissue Disorders: Uncommon: Rash, pruritus, urticaria. Rare: Angioedema. Very Rare: Stevens-Johnson syndrome.

Reproductive System and Breast Disorders: Common: Ejaculation disorders. Very Rare: Priapism.

General Disorders and Administration Site Conditions: Uncommon: Asthenia.

As with other α -blockers, drowsiness, blurred vision, dry mouth or oedema can occur.

During cataract surgery, a small pupil situation, known as IFIS, has

been associated with therapy of tamsulosin during post-marketing surveillance (see Precautions).

Post-Marketing Experience: In addition to the adverse reactions previously mentioned, atrial fibrillation, arrhythmia, tachycardia and dyspnoea have been reported in association with tamsulosin use. Because these spontaneously reported events are from the worldwide post-marketing experience, the frequency of events and the role of tamsulosin in their causation cannot be reliably determined.

Overdose and Treatment

Symptoms: Overdosage with tamsulosin hydrochloride can potentially result in severe hypotensive effects, dizziness and malaise. Severe hypotensive effects have been observed at different levels of overdosing.

Treatment: In case of acute hypotension occurring after overdosage, cardiovascular support should be given. Blood pressure can be restored and heart rate brought back to normal by lying the patient down. If this does not help, then volume expanders, and when necessary, vasopressors could be employed. Renal function should be monitored and general supportive measures applied. Dialysis is unlikely to be of help, as tamsulosin is very highly bound to plasma proteins.

Measures eg, emesis, can be taken to impede absorption. When large quantities are involved, gastric lavage can be applied, and activated charcoal and an osmotic laxative eg, sodium sulfate can be administered.

Storage Conditions

Store below 30°C.

Dosage Forms and Packaging Available

Available in blister packages of 30 (3 x 10) and 100 (10x10) tablets.

Not all pack size is marketed.

Name and Address of Manufacturer

Apotex Inc.

150 Signet Drive Toronto, Ontario Canada M9L 1T9

Name and Address of Marketing Authorization Holder

Pharmaforte (M) Sdn Bhd

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Date of Revision of Package Insert

24 June 2022

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